County-Level Food Insecurity and Chronic Disease Burden in the United States: Identifying Priority Counties for Intervention

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1 Abstract

Food insecurity and chronic disease frequently occur in the same U.S. communities, yet public-health planning often treats nutrition-related social need and chronic disease prevention as separate problems. This study examined county-level associations between adjusted food insecurity prevalence and four chronic disease outcomes—coronary heart disease, diabetes, obesity, and frequent poor mental health—and used these outcomes to construct a standardized chronic disease burden index. The analysis combined descriptive statistics, correlation analysis, ordinary least-squares regression, priority-county classification, mapping, sensitivity analysis, and Moran's I spatial autocorrelation.

The complete analytic sample included 2,299 counties. Mean food insecurity was 17.30%, with values ranging from 5.9% to 46.8%. Food insecurity was positively correlated with all four outcomes, most strongly with diabetes ($r = 0.812$), frequent poor mental health ($r = 0.625$), and obesity ($r = 0.553$). In adjusted models, food insecurity remained positively associated with coronary heart disease, diabetes, obesity, and the composite burden index, although the frequent-poor-mental-health coefficient reversed sign after adjustment in a highly collinear model. Using 75th-percentile cutoffs for both food insecurity and disease burden, 408 counties were classified as priority counties. These counties were concentrated in the Deep South, lower Mississippi Valley, Appalachia, and selected rural and tribal regions. Sensitivity analyses gave similar geographic patterns, and Moran's I showed significant spatial clustering.

The findings are ecological and cross-sectional and therefore do not establish individual-level causation. They do, however, provide a practical framework for identifying counties where nutrition assistance, chronic disease prevention, transportation support, and health-care access efforts may be especially valuable.

2 Introduction

2.1 Background and Motivation

Food insecurity is limited or uncertain access to adequate food. It reflects more than food availability alone: it is closely tied to income, transportation, housing stability, public assistance, and the local food environment. These conditions can influence diet quality, psychological stress, medication adherence, and the tradeoffs households make between food, health care, utilities, and transportation (ODPHP, 2023; Seligman et al., 2010; Laraia, 2013; Thomas et al., 2021).

Chronic diseases such as diabetes, obesity, coronary heart disease, and poor mental health are also shaped by place. County-level differences in economic conditions, health-care access, transportation, social isolation, and health behaviors can produce large geographic disparities in disease burden. The CDC PLACES program provides local estimates of chronic disease, health behavior, disability, health status, and social needs, making it possible to examine these patterns below the state level (Greenlund et al., 2022; CDC, 2025a,b).

The practical question motivating this project is therefore not simply whether food insecurity is associated with chronic disease, but *where the two problems overlap most strongly.* Counties facing both high food insecurity and high chronic disease burden may benefit from coordinated interventions rather than separate nutrition and disease-prevention programs.

2.2 Related Evidence

Prior research supports several pathways linking food insecurity with chronic disease. Food-insecure households may rely more heavily on inexpensive, calorie-dense foods, experience irregular eating patterns, and have greater difficulty sustaining disease-management routines. These processes are relevant to obesity and diabetes (Laraia, 2013; Seligman et al., 2010). Food insecurity may also contribute to cardiovascular risk through poorer diet quality, stress, medication tradeoffs, and reduced access to preventive care (Thomas et al., 2021). For mental health, uncertainty about meeting basic needs can create persistent psychological strain (Myers, 2020).

Most of this literature focuses on individuals or households. This project instead uses a county-level framework designed for public-health planning. The contribution is to combine multiple chronic disease outcomes, a composite burden index, threshold-based priority classification, maps, sensitivity analysis, and spatial autocorrelation in one coherent county-level analysis.

2.3 Research Questions

The analysis addresses five questions:

1. How do county-level food insecurity and chronic disease outcomes vary across the United States?
2. How strongly is food insecurity associated with coronary heart disease, diabetes, obesity, and frequent poor mental health?
3. Does food insecurity remain associated with these outcomes after accounting for related social-needs and behavioral variables?
4. Which counties have jointly high food insecurity and high chronic disease burden, and where are they located?
5. Are the priority-county results robust to alternative thresholds, and do they show measurable spatial clustering?

2.4 Conceptual Framework

The conceptual framework treats food insecurity as one component of a broader social-risk environment. County social conditions may contribute to food insecurity, which may then operate through diet quality, chronic stress, and barriers to medication or care. Chronic disease may also worsen financial strain, creating a reinforcing cycle. Because the data are cross-sectional and county-level, these pathways are interpreted as plausible mechanisms rather than causal effects.

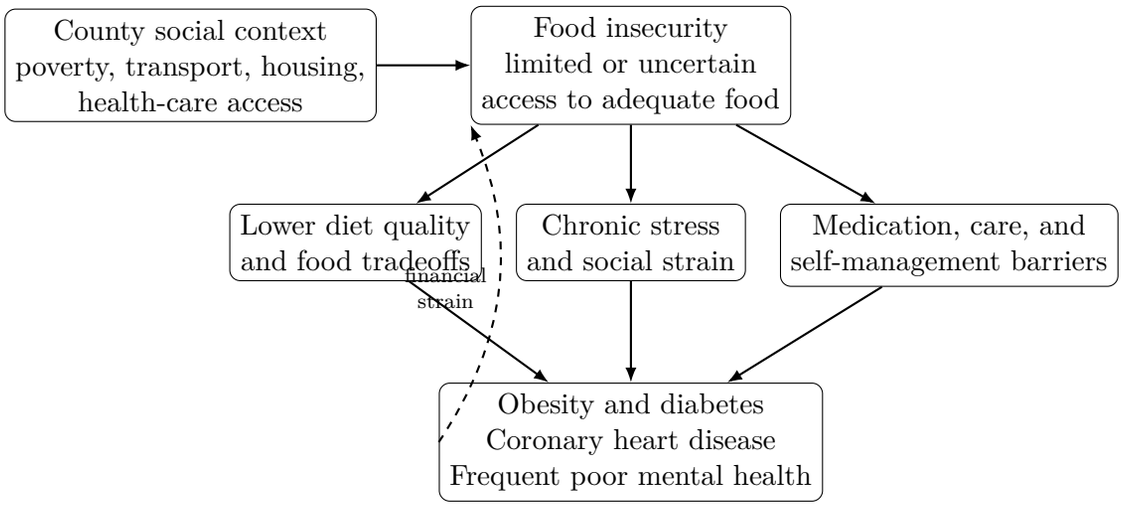


Figure 1: Conceptual framework linking county social context, food insecurity, intermediate pathways, and chronic disease outcomes. The dashed arrow indicates a possible feedback pathway in which chronic disease worsens financial strain.

3 Data and Methods

3.1 Data Sources and Analytic Sample

The primary analytic file combines county-level chronic disease, health behavior, and health-related social-needs measures. Health measures were drawn from CDC PLACES, which uses BRFSS, U.S. Census, and American Community Survey information with small-area estimation methods to generate local estimates (CDC, 2024a, 2025a,b). The food insecurity measure in the assembled analytic file is an adjusted county-level prevalence estimate. In the broader data ecosystem, local food insecurity estimates are commonly produced by Feeding America's *Map the Meal Gap* project (Feeding America, 2025a,b).

The raw county file contained 3,143 county or county-equivalent records. The main analysis used a fixed complete-case sample of 2,299 counties with non-missing values for the variables included in the correlation, regression, and priority-county analyses. Using the same counties across analyses improves comparability among models and figures.

3.2 Variables

Food insecurity is the main exposure. The four chronic disease outcomes are coronary heart disease, diabetes, obesity, and frequent poor mental health. Adjusted models additionally include housing insecurity, lack of reliable transportation, food stamp/SNAP use, loneliness, current smoking, and physical inactivity. These variables represent related social needs or behavioral factors that may overlap with both food insecurity and chronic disease.

Table 1: Core variables and their roles in the analysis

Variable	Role	Interpretation
Food insecurity, adjusted prevalence	Main predictor	Estimated percentage of county residents experiencing food insecurity; higher values indicate greater food-related social need.
Coronary heart disease	Outcome	Estimated adult prevalence of coronary heart disease.
Diabetes	Outcome	Estimated adult prevalence of diagnosed diabetes.
Obesity	Outcome	Estimated adult prevalence of obesity.
Frequent poor mental health	Outcome	Estimated percentage of adults reporting poor mental health for at least 14 of the past 30 days.
Chronic disease burden index	Composite outcome	Average of standardized diabetes, obesity, coronary heart disease, and frequent poor mental health values.
Housing insecurity; transportation barriers; SNAP use; loneliness	Covariates	Related county-level social-needs measures.
Current smoking; physical inactivity	Covariates	County-level behavioral risk measures.

3.3 Chronic Disease Burden Index

To summarize multiple outcomes in a single measure, each chronic disease outcome was standardized across counties. For county c and outcome k,

$$z_{kc} = \frac{Y_{kc} - \bar{Y}_k}{s_k}, \tag{1}$$

where $\bar{Y}_k$ and s_k are the sample mean and standard deviation for outcome k. The chronic disease burden index was then defined as

$$B_c = \frac{1}{4}\left(z_{\text{diabetes},c} + z_{\text{obesity},c} + z_{\text{CHD},c} + z_{\text{MHLTH},c}\right). \tag{2}$$

Positive values indicate above-average combined burden, while negative values indicate below-average burden relative to the analytic sample.

3.4 Association and Regression Models

Pairwise relationships were summarized with Pearson's correlation coefficient. For each chronic disease outcome, the unadjusted regression model was

$$Y_c = \beta_0 + \beta_1 F_c + \varepsilon_c, \tag{3}$$

where F_c denotes food insecurity prevalence. The adjusted model was

$$Y_c = \beta_0 + \beta_1 F_c + \boldsymbol{\gamma}^\top \mathbf{X}_c + \varepsilon_c, \tag{4}$$

where $\mathbf{X}_c$ contains housing insecurity, transportation barriers, SNAP use, loneliness, current smoking, and physical inactivity. The same structure was used with B_c as the outcome.

Because several social-needs variables are extremely highly correlated, adjusted coefficients are interpreted as *conditional associations*, not isolated causal effects. This distinction is especially important for the frequent-poor-mental-health model and for coefficients that change sign after adjustment.

3.5 Priority Counties, Sensitivity Analysis, and Spatial Clustering

Priority counties were defined as those at or above the 75th percentile for both food insecurity and the chronic disease burden index:

$$P_c = \mathbf{1}\left\{F_c \geq Q_{0.75}(F) \text{ and } B_c \geq Q_{0.75}(B)\right\}. \tag{5}$$

In the analytic sample, these thresholds were approximately 20.05% for food insecurity and 0.53 for the burden index. Sensitivity analyses repeated the classification at the 70th, 80th, and 90th percentiles and also used a fixed 20% food insecurity threshold paired with the 75th-percentile burden threshold.

Spatial clustering was summarized using Moran's I with a 6-nearest-neighbor matrix based on county centroids. Permutation-based p-values used 999 random permutations. Positive Moran's I indicates that counties with similar values tend to be geographically near one another.

4 Results

4.1 Descriptive Statistics

Food insecurity averaged 17.30% across the 2,299 counties and ranged from 5.9% to 46.8%. Among the chronic disease outcomes, obesity had the highest mean prevalence (37.88%), followed by frequent poor mental health (17.07%), diabetes (13.63%), and coronary heart disease (7.83%). The burden index had a mean near zero by construction and ranged from -2.34 to 2.94, indicating substantial cross-county variation in combined disease burden.

Table 2: Descriptive statistics for the main analytic variables

Variable	N	Mean	SD	Min	Max
Food insecurity (%)	2299	17.30	6.10	5.9	46.8
Diabetes (%)	2299	13.63	2.79	4.9	27.1
Obesity (%)	2299	37.88	4.61	17.2	52.9
Coronary heart disease (%)	2299	7.83	1.48	3.0	13.3
Frequent poor mental health (%)	2299	17.07	1.98	11.8	24.5
Housing insecurity (%)	2299	13.97	4.26	5.9	33.6
Transportation barriers (%)	2299	9.45	2.86	4.1	26.8
SNAP use (%)	2299	14.05	5.93	3.6	46.6
Loneliness (%)	2299	34.19	2.80	25.7	43.8
Current smoking (%)	2299	15.96	3.34	6.4	36.6
Physical inactivity (%)	2299	28.67	5.35	13.5	47.9

4.2 Correlation Patterns

Food insecurity was positively correlated with all four chronic disease outcomes. The association was strongest for diabetes ($r = 0.812$), followed by frequent poor mental health ($r = 0.625$), obesity ($r = 0.553$), and coronary heart disease ($r = 0.450$). Food insecurity was also extremely highly correlated with housing insecurity ($r = 0.985$) and transportation barriers ($r = 0.983$), an important source of multicollinearity in adjusted models.

Table 3: Pearson correlations with food insecurity

Variable	Correlation
Diabetes	0.812
Obesity	0.553
Coronary heart disease	0.450
Frequent poor mental health	0.625
Housing insecurity	0.985
Transportation barriers	0.983
SNAP use	0.903
Loneliness	0.735
Current smoking	0.672
Physical inactivity	0.779

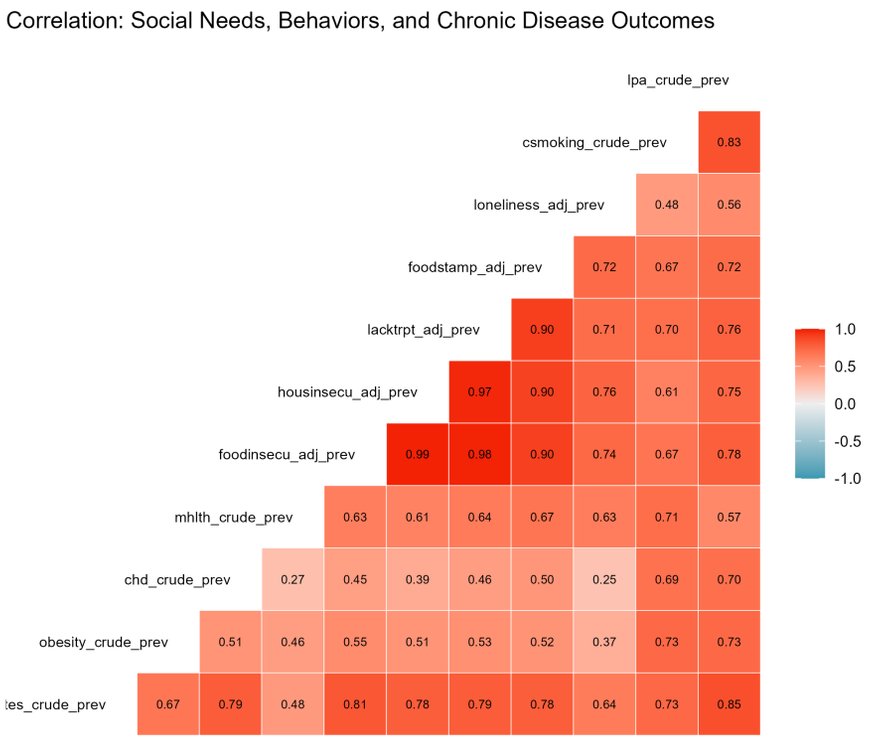
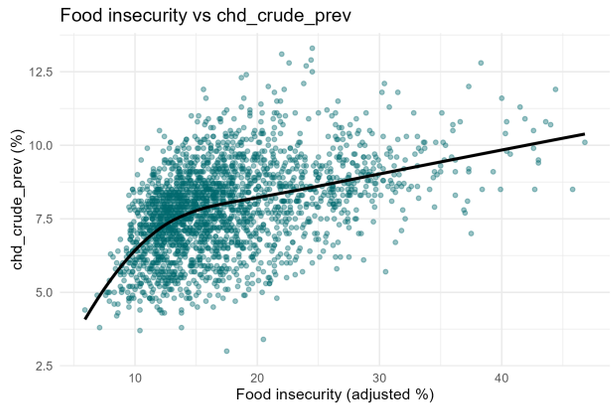


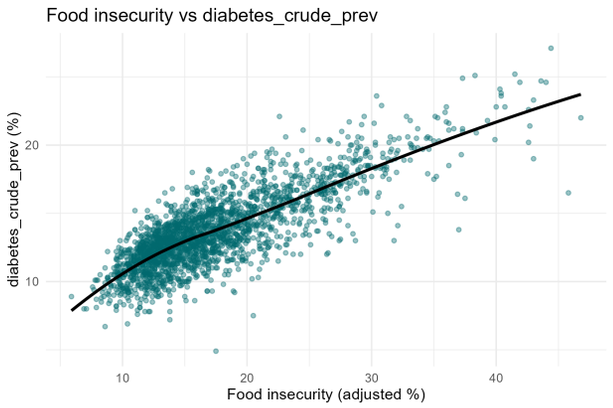
Figure 2: Correlation heatmap of social needs, behavioral risk factors, and chronic disease outcomes. The strongest positive relationships occur among food insecurity, housing insecurity, transportation barriers, diabetes, smoking, and physical inactivity.

4.3 Bivariate Relationships with Food Insecurity

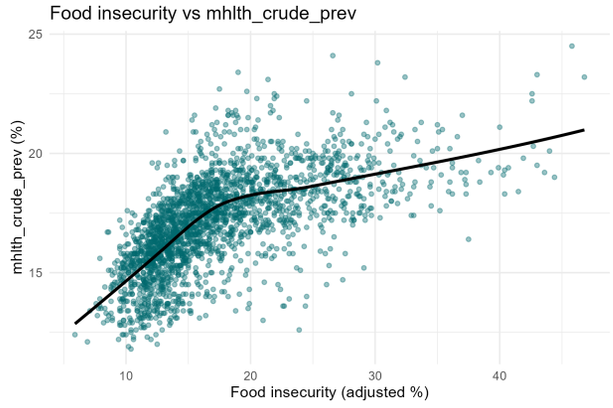
The scatterplots in Figure 3 show positive bivariate relationships between food insecurity and each chronic disease outcome. Diabetes exhibits the clearest upward pattern. Obesity also rises substantially with food insecurity, although the pattern is nonlinear. Coronary heart disease is positively associated with food insecurity but has greater dispersion. Frequent poor mental health also rises with food insecurity before adjustment for other social-needs and behavioral variables.



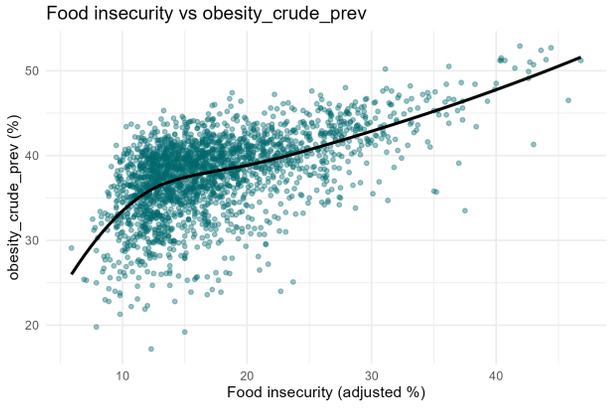
(a) Coronary heart disease



(b) Diabetes



(c) Frequent poor mental health



(d) Obesity

Figure 3: County-level bivariate relationships between adjusted food insecurity prevalence and chronic disease outcomes.

4.4 Regression Results

In unadjusted models, a one-percentage-point increase in food insecurity was associated with higher prevalence of every chronic disease outcome. After adjustment, food insecurity remained positively associated with coronary heart disease, diabetes, and obesity. The adjusted association was largest for obesity (0.674 percentage points), followed by diabetes (0.318) and coronary heart disease (0.056).

The frequent-poor-mental-health coefficient changed from positive in the unadjusted model to slightly negative in the adjusted model. Because food insecurity is almost perfectly correlated with housing insecurity and transportation barriers and is also strongly correlated with several other covariates, this reversal should not be interpreted as a protective effect. It reflects the narrow meaning of a coefficient estimated after holding several overlapping social-risk measures constant.

Table 4: Food insecurity coefficients from unadjusted and adjusted chronic disease models

Outcome	Model	Estimate	95% CI	p-value
Coronary heart disease	Unadjusted	0.110	(0.101, 0.118)	< 0.001
	Adjusted	0.056	(0.007, 0.106)	0.025
Diabetes	Unadjusted	0.372	(0.361, 0.383)	< 0.001
	Adjusted	0.318	(0.251, 0.386)	< 0.001
Obesity	Unadjusted	0.418	(0.393, 0.444)	< 0.001
	Adjusted	0.674	(0.521, 0.827)	< 0.001
Frequent poor mental health	Unadjusted	0.203	(0.193, 0.214)	< 0.001
	Adjusted	-0.065	(-0.128, -0.003)	0.040

4.5 Composite Chronic Disease Burden

Food insecurity was positively associated with the chronic disease burden index in both model specifications. In the unadjusted model, each one-percentage-point increase in food insecurity corresponded to a 0.100-unit increase in the burden index. After adjustment, the coefficient was 0.066 and remained statistically significant.

Table 5: Regression of chronic disease burden index on food insecurity

Model	Predictor	Estimate	95% CI	p-value
Unadjusted	Food insecurity	0.100	(0.097, 0.104)	< 0.001
Adjusted	Food insecurity	0.066	(0.052, 0.081)	< 0.001
Adjusted	Housing insecurity	-0.063	(-0.082, -0.044)	< 0.001
Adjusted	Transportation barriers	-0.087	(-0.113, -0.061)	< 0.001
Adjusted	SNAP use	0.037	(0.032, 0.041)	< 0.001
Adjusted	Loneliness	0.021	(0.015, 0.027)	< 0.001
Adjusted	Current smoking	0.106	(0.098, 0.114)	< 0.001
Adjusted	Physical inactivity	0.057	(0.052, 0.062)	< 0.001

4.6 Priority Counties

Using the 75th-percentile thresholds, 408 of 2,299 counties (17.7%) were classified as priority counties. These counties had food insecurity of at least approximately 20.05% and a burden index of at least approximately 0.53. Figure 4 shows that they occupy the upper-right portion of the food-insecurity–burden relationship.

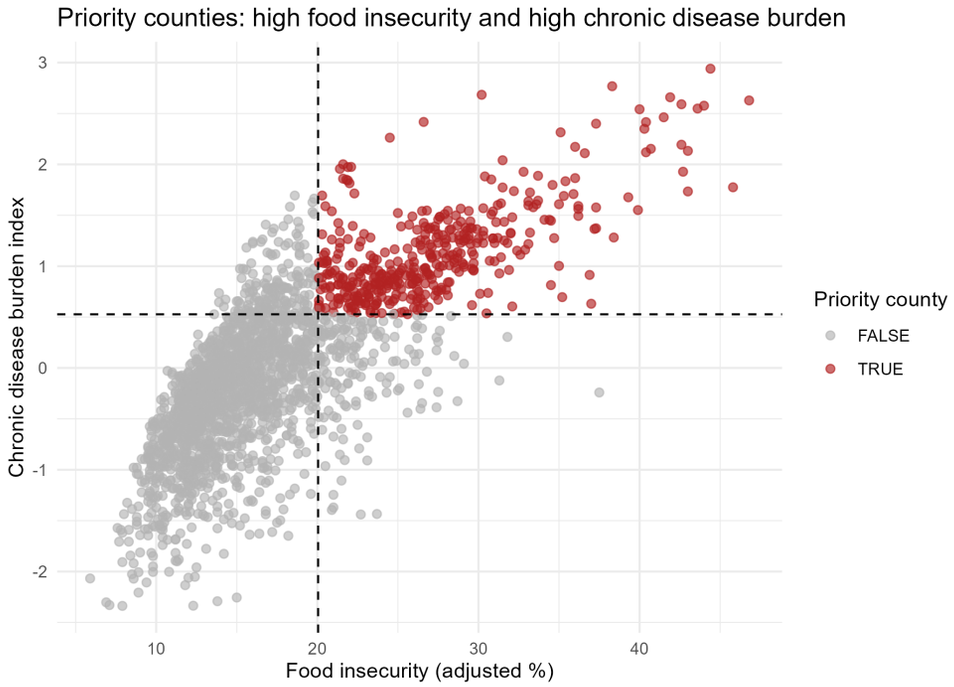


Figure 4: Priority counties with high food insecurity and high chronic disease burden. Red points indicate counties at or above the 75th percentile on both measures.

The highest-burden priority counties were concentrated in Alabama, Louisiana, Mississippi, West Virginia, Georgia, and nearby states. Greene County, Alabama had the highest burden index among the listed priority counties (2.94), followed by Tensas Parish, Louisiana (2.77) and McDowell County, West Virginia (2.68).

Table 6: Top 10 priority counties ranked by chronic disease burden index

Rank	County	Food insec.	Burden	Diab.	Obesity	CHD	Poor MH
1	Greene, AL	44.4	2.94	27.1	52.7	11.9	19.0
2	Tensas, LA	38.3	2.77	25.1	48.4	12.8	19.7
3	McDowell, WV	30.2	2.68	21.6	46.2	11.8	23.8
4	Perry, AL	41.9	2.66	24.6	52.9	10.9	19.8
5	East Carroll, LA	46.8	2.63	22.0	51.2	10.1	23.2
6	Madison, LA	42.6	2.59	22.6	49.9	10.5	22.5
7	Bullock, AL	44.0	2.58	24.6	51.3	10.7	20.1
8	Humphreys, MS	43.6	2.55	24.7	52.4	10.8	19.2
9	Claiborne, LA	40.0	2.54	22.8	48.5	11.6	21.1
10	Sharkey, MS	41.5	2.46	25.2	50.3	11.3	18.4

Georgia had the largest number of priority counties (71), followed by Mississippi (51), Louisiana (46), Alabama (40), Arkansas (40), and Oklahoma (31). These state counts align with the geographic clustering visible in the maps.

4.7 Geographic Patterns

The bivariate map in Figure 5 shows a clear geographic structure. Counties with high values of both food insecurity and chronic disease burden are concentrated in the Deep South, lower

Mississippi Valley, parts of Appalachia, selected rural areas in the Southwest and Northern Plains, and parts of Alaska.

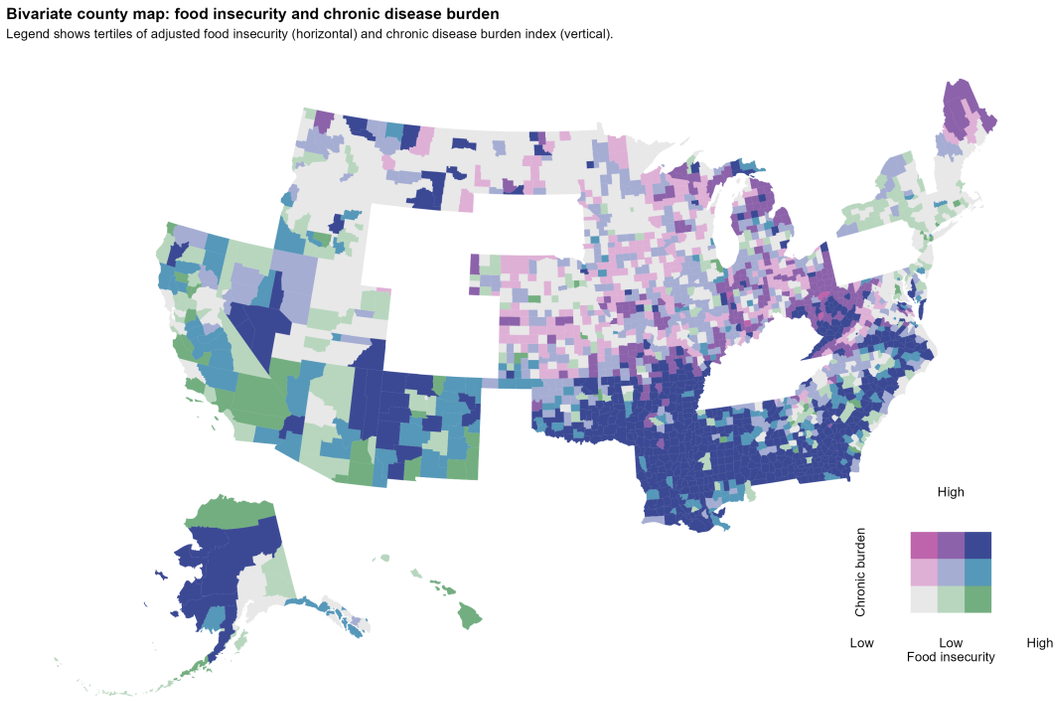


Figure 5: Bivariate county map of adjusted food insecurity and chronic disease burden. The high-high category identifies areas where both dimensions are elevated.

The binary priority map in Figure 6 makes the same pattern easier to interpret for planning: the largest contiguous concentration is in the South, with additional clusters in Appalachia and selected rural or tribal regions.

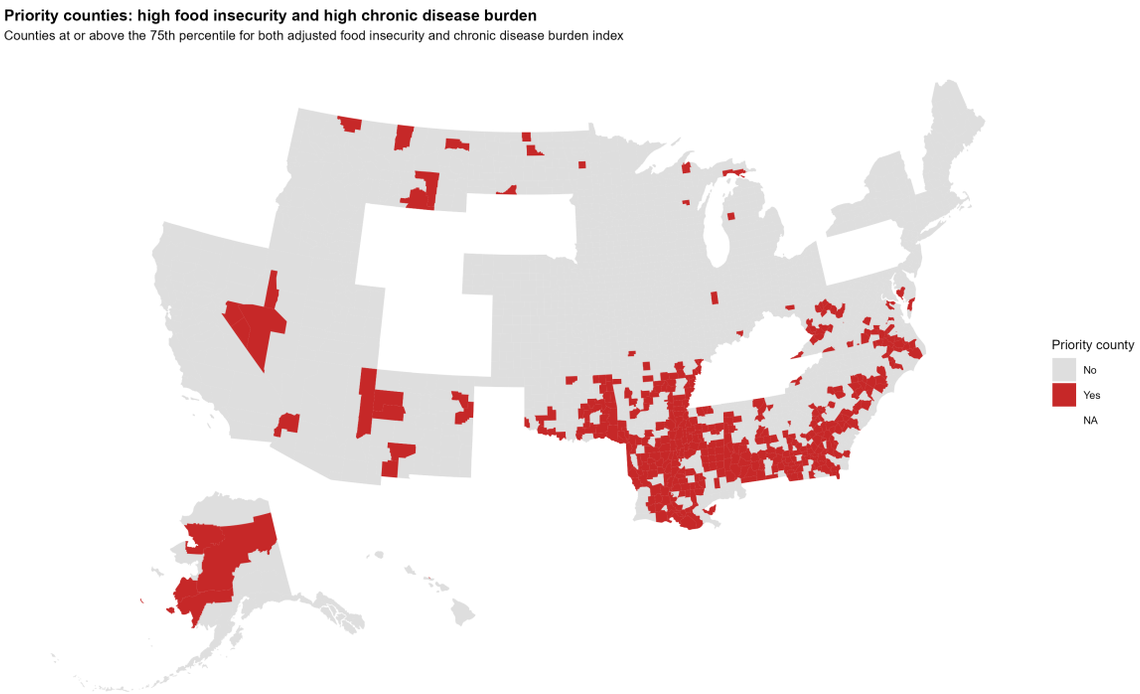


Figure 6: Priority county map. Red counties are at or above the 75th percentile for both food insecurity and chronic disease burden.

4.8 Sensitivity Analysis and Spatial Autocorrelation

The number of priority counties changed as the threshold became more or less restrictive, but the classification remained substantively similar. At the 70th percentile, 483 counties were classified as high-high; at the 80th percentile, 319; and at the 90th percentile, 152. A fixed 20% food insecurity cutoff paired with the 75th-percentile burden threshold identified 411 counties—almost identical to the 408 counties from the main definition.

Table 7: Sensitivity of the priority-county classification to alternative thresholds

Definition	Food threshold	Burden threshold	Counties	Percent
70th percentile	18.9	0.376	483	21.0%
75th percentile	20.0	0.527	408	17.7%
80th percentile	21.5	0.679	319	13.9%
90th percentile	26.2	1.040	152	6.6%
20% food insecurity + 75th burden	20.0	0.527	411	17.9%

Moran's I indicated strong and statistically significant spatial clustering for food insecurity ($I = 0.643$), chronic disease burden ($I = 0.630$), and the priority-county flag ($I = 0.494$), with all permutation-based p-values below 0.001. The maps therefore reflect measurable geographic clustering rather than purely random spatial variation.

Table 8: Moran's I spatial autocorrelation using 6-nearest-neighbor county centroid weights

Variable	Moran's I	Expected I	Statistic	p-value
Food insecurity	0.643	-0.000435	55.4	< 0.001
Chronic disease burden index	0.630	-0.000435	54.3	< 0.001
Priority county flag	0.494	-0.000435	42.5	< 0.001

5 Discussion

5.1 Main Findings

Three findings stand out. First, counties with higher food insecurity generally had higher chronic disease burden. The strongest bivariate relationship was with diabetes, but positive patterns were also observed for obesity, coronary heart disease, and frequent poor mental health. Second, food insecurity remained positively associated with diabetes, obesity, coronary heart disease, and the composite burden index after adjustment for several overlapping social and behavioral variables. Third, counties with simultaneously high food insecurity and high disease burden were geographically concentrated rather than randomly distributed.

These results support the interpretation of food insecurity as both a nutrition-related hardship and a marker of broader structural disadvantage. The very high correlations between food insecurity, housing insecurity, and transportation barriers show that these risks often occur together. Consequently, the adjusted regression coefficients should not be read as clean causal estimates of separate mechanisms.

5.2 Public Health Interpretation

The findings are consistent with several mechanisms described in the literature. Food insecurity may increase reliance on inexpensive, calorie-dense foods and make it harder to maintain stable diets, which can complicate diabetes prevention and management. Financial strain can also interfere with medication adherence and routine care. Chronic stress may further contribute to cardiovascular and mental-health burden. These pathways are plausible at the population level, but the present analysis cannot establish which pathway is responsible for a particular county's pattern.

The priority-county framework is useful because it shifts the emphasis from a single national association to local planning. Counties with both high food insecurity and high chronic disease burden may be logical candidates for coordinated programs involving SNAP outreach, food banks, produce prescription programs, diabetes prevention, cardiovascular risk management, community health workers, transportation support, and stronger health-care referral systems.

5.3 Strengths and Limitations

A major strength of the project is its national county-level scope. The analysis considers multiple chronic disease outcomes, combines them into a transparent standardized burden index, and integrates regression, mapping, sensitivity analysis, and spatial statistics. The agreement between the main threshold definition and the fixed 20% food insecurity sensitivity check strengthens the practical interpretation of the priority-county classification.

Several limitations are important. First, the analysis is cross-sectional and ecological, so it cannot establish temporal ordering or individual-level causation. Second, county averages can conceal substantial within-county inequality. Third, CDC PLACES estimates are model-based small-area estimates and therefore contain uncertainty not fully represented by ordinary regression standard errors. Fourth, the analytic file does not include several potentially important county-level confounders, including poverty, income, unemployment, education, racial and ethnic composition, uninsured rate, and rural–urban status. Finally, the extreme correlation among several social-needs measures creates multicollinearity and makes adjusted coefficients sensitive to model specification.

5.4 Future Work

The most useful extensions would be to add county-level socioeconomic and health-care context, incorporate the CDC/ATSDR Social Vulnerability Index, merge USDA Food Access Research Atlas measures, and examine whether associations differ by region or rural–urban status. A longitudinal county panel would improve temporal interpretation, while tract-level analyses could reveal hotspots hidden by county averages. Spatial analysis could also be extended from centroid-based nearest-neighbor weights to county-boundary contiguity weights and local cluster statistics such as LISA or Getis–Ord G_i^*.

6 Conclusion

County-level food insecurity and chronic disease burden are closely connected in the United States. Higher food insecurity is associated with higher diabetes, obesity, coronary heart disease, frequent poor mental health, and overall chronic disease burden at the bivariate level, while several associations remain after statistical adjustment. The counties facing the highest combined

burden are concentrated in identifiable geographic regions, especially the Deep South, lower Mississippi Valley, Appalachia, and selected rural and tribal areas.

The principal contribution of this project is therefore practical rather than causal. It provides a reproducible way to identify counties where nutrition-related social need and chronic disease burden overlap and where coordinated interventions may be especially valuable. The results should be used as a targeting and hypothesis-generating tool, not as evidence that food insecurity alone causes chronic disease.

Appendix

A. Interpretation of Adjusted Models

Food insecurity was correlated at $r = 0.985$ with housing insecurity and $r = 0.983$ with transportation barriers. In such a highly collinear setting, adjusted coefficients answer a narrow question: what is the association with food insecurity when other closely overlapping county-level social-risk measures are held fixed? This is why coefficients may shrink or change sign after adjustment even when the overall bivariate association is positive.

B. Spatial Analysis Note

Moran's I used 6-nearest-neighbor county centroid weights. This is a practical method when polygon adjacency files are not used, but future work could use county shapefiles with queen- or rook-contiguity weights. Local Indicators of Spatial Association (LISA) or Getis–Ord G_i^* could then identify specific high-high clusters more formally.

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